Remote Work across Jobs, Companies, and Space

Stephen Hansen, Peter John Lambert, Nicholas Bloom, Steven J. Davis, Raffaella Sadun, and Bledi Taska

NBER Working Paper 31007, 2026

Prepared for ifo Summer Course 2026, by Benjamin W. Arold

Motivation & Research Question

- COVID-19 triggered an abrupt, large-scale, and enduring shift to remote work; by 2025 about one-quarter of full U.S. workdays are done from home.
- Most evidence comes from worker/employer surveys or occupation-level 'teleworkability' scores -- limited in scale, granularity, and frequency.
- Question: how large, how persistent, and how UNEVEN is the shift to advertised remote work across countries, occupations, cities, and firms?
- Approach: read the full TEXT of >500 million job vacancy postings and detect which ones explicitly offer hybrid or fully-remote work.
- Key challenge is a measurement problem -- turning messy job-ad text into an accurate remote-work classification at scale.

Data

- Near-universe of online vacancy postings from Lightcast (formerly Emsi Burning Glass), scraped from 50,000+ job boards, aggregators, and employer sites.
- Five English-speaking countries: United States, United Kingdom, Australia, Canada, and New Zealand.
- January 2014 - November 2025: 550+ million postings, 12.5 million employers, ~62,000 cities (5% sample pre-2019, full universe after).
- Each posting has plain text plus date, employer, occupation, industry, and location; median posting length is 347 words.
- Baseline series re-weight each country-month to the 2024 U.S. occupation mix to strip out compositional differences.

The Measurement Problem: Why Dictionaries Fail

- Standard text-as-data approach: flag a posting positive if it contains keywords like 'remote', 'work from home', 'telework'.
  - False positives: 'deep-sea wells in remote locations', company 'home offices', 'work from home care facilities', 'requires a Home Office work permit'.
  - Structural break: after early 2020 many ads add a field that explicitly states remote work is NOT allowed -- naive keyword matching reads it as positive.
- False negatives: firms describe remote work in many creative ways no fixed keyword list can capture.
- Result: dictionaries have high error rates that vary over time and across occupations -- meaning depends on CONTEXT, not word presence.
- 'Some deep-sea wells we operate are in remote locations' vs. 'We are pleased to offer remote work' -- same word, opposite meaning.

Method: Building the RWO Classifier (DistilBERT)

- Foundation model: DistilBERT -- a compact 66M-parameter Transformer (vs 110M for BERT) that reads text via self-attention, letting each word's meaning depend on its context.
- Step 1 -- Unsupervised fine-tuning: re-train DistilBERT to predict masked words in ~900k job-posting sequences, so representations fit the language of job ads.
- Step 2 -- Human labels: split each posting into ~45-word sequences; three Amazon Mechanical Turk readers label 10,000 sequences (=30,000 labels); pairwise agreement >90%.
- Step 3 -- Supervised fine-tuning: train the model to predict the human labels -> the 'Remote Work in job Openings' (RWO) classifier.
- Apply out-of-sample to ~1.6 billion sequences; a posting is positive if ANY sequence is positive. Fully open-source; ~94 GPU-hours, ~$3,200 total.

Method: Attention Weights Read Context, Not Keywords

Table 3: Attention Weights from the RWO large language model, as compared to the Dictionary Keywords

RWO View:	Dictionary View:
Schedule: ** 10 Hour Shift ** 8 Hour Shift Work remotely: ** No	Schedule: ** 10 Hour Shift ** 8 Hour Shift Work remotely: ** No
We are looking for a Deputy Home Manager with domiciliary care experience to join our company. You will work from home care facilities with a strong track record of quality service.	We are looking for a Deputy Home Manager with domiciliary care experience to join our company. You will work from home care facilities with a strong track record of quality service.
We encourage our people to explore new ways of working - including part-time, job-share or working from different kinds of locations, including their home. Everyone can ask about it.	We encourage our people to explore new ways of working - including part-time, job-share or working from different kinds of locations, including their home. Everyone can ask about it.

Note: Left (RWO): darker shading = higher attention weight. The model learns to down-weight 'remote/home' in negations and irrelevant contexts and up-weight genuine offers. Right (Dictionary): a keyword hit fires regardless of surrounding meaning, generating the false positives shown. (Table 3)

RWO Outperforms Every Alternative Method

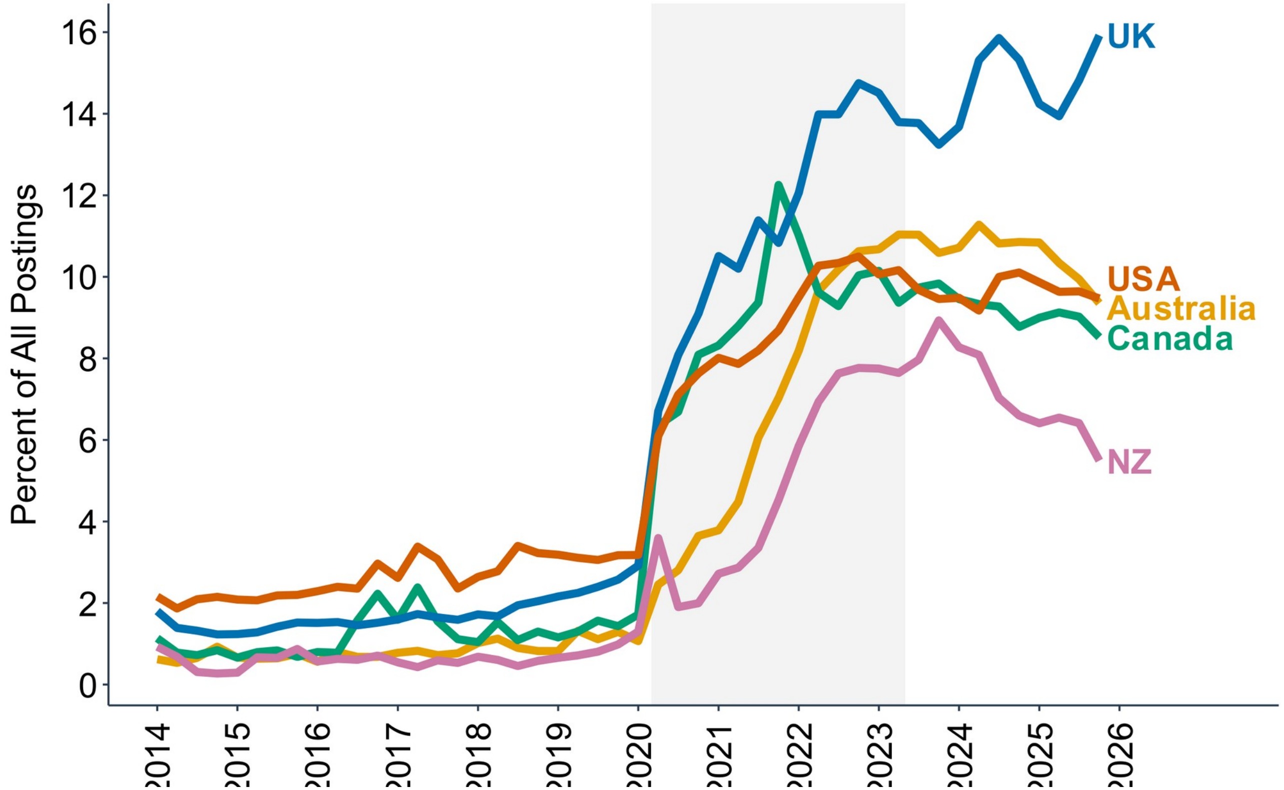
Table 4: RWO Outperforms Other Classification Methods

	Audit Sample			Approximate Random Sample		
	(1)	(2)	(3)	(4)	(5)	(6)
	Error Rate	Precision	F1 Score	Error Rate	Precision	F1 Score
All Zero	.28	.00	.00	.03	.00	.00
Dictionary	.16	.68	.74	.14	.15	.25
Dictionary w/ Negation	.12	.82	.78	.07	.28	.40
Logistic Regression	.11	.81	.81	.07	.26	.40
Logistic Regression w/ Negation	.08	.87	.85	.05	.36	.50
GPT-4o	.03	.95	.94	.02	.59	.73
Claude Sonnet 4	.04	.89	.93	.05	.38	.55
RWO (Generic English)	.03	.95	.95	.02	.66	.78
RWO (Baseline)	.02	.97	.97	.01	.75	.85

Note: Test-set performance. RWO (Baseline) reaches a 0.02 error rate and 0.97 F1 -- the dictionary's error rate is 8x higher. On the harder 'random sample' (columns 4-6, ~3% positive) RWO's F1 (0.85) beats GPT-4o (0.73) and Claude Sonnet 4 (0.55), while running at a fraction of frontier-model cost. Unsupervised fine-tuning matters (Generic English F1 falls to 0.78). (Table 4)

Results: A Sharp, Persistent Rise across Countries

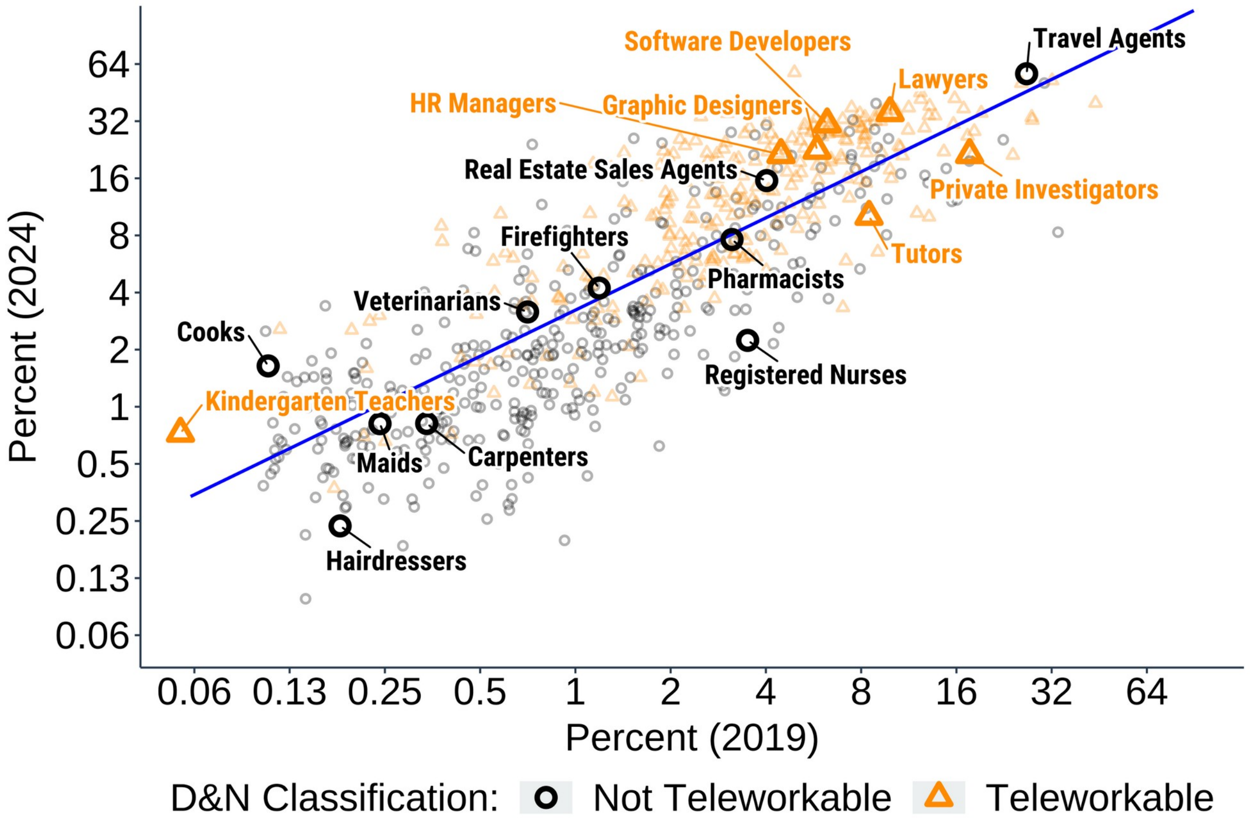
Figure 1: Vacancy Postings that Explicitly Offer Hybrid or Fully Remote Work Rose Sharply and Persistently



Note: Weighted share of postings offering >=1 remote day/week, 2014-2025. Roughly doubled Feb->Apr 2020, then kept climbing rather than reverting. By late 2025: ~16% in the UK and a 9-10% 'new normal' in the US, Canada, and Australia -- a 3x (US) to 5x+ (other countries) rise since 2019. NZ partially reverts from its 2024 peak. (Figure 1)

Heterogeneity across Occupations (and Space)

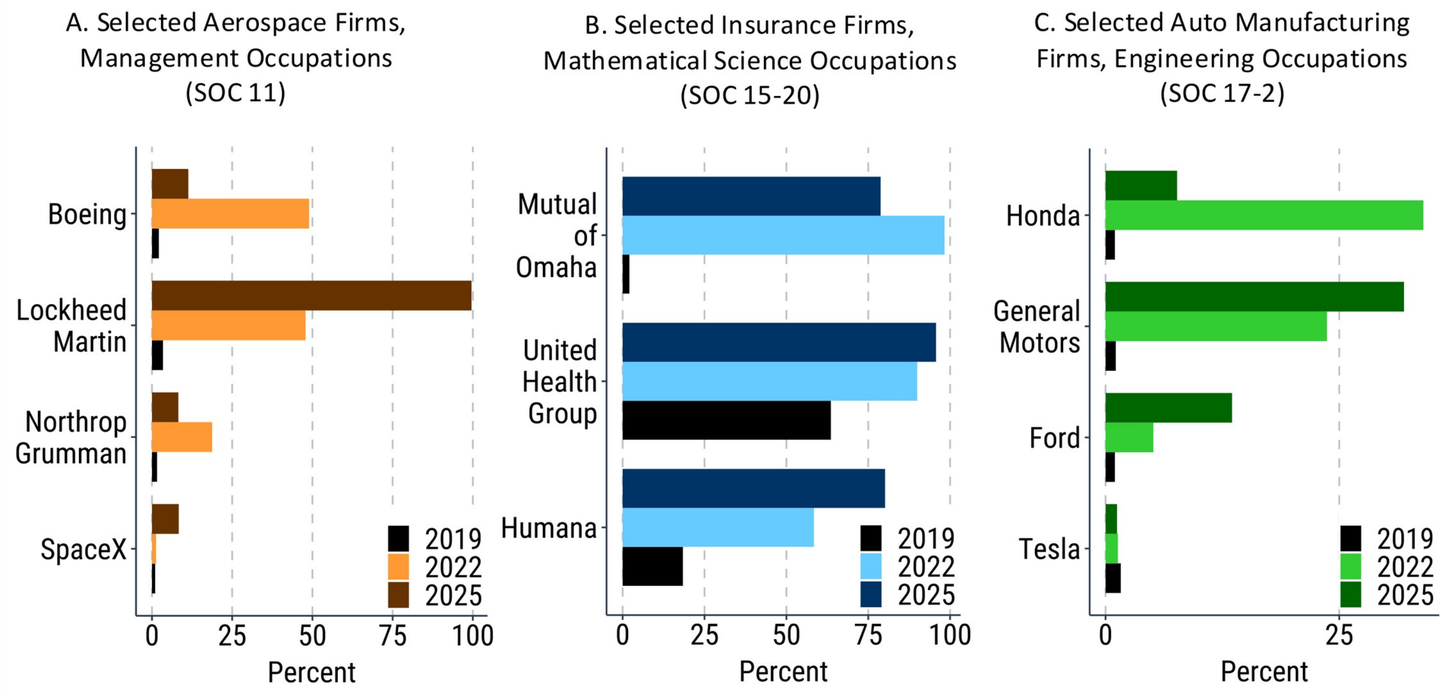
Figure 3: The Share of Vacancy Postings that Explicitly Offer Hybrid or Fully Remote Work Rose in Almost Every Occupation, U.S. Data



*Note: 719 U.S. O*NET occupations, 2019 vs 2024 (log-log). Almost all rose; 2019 shares strongly predict 2024 (R2 = 0.80). D&N 'teleworkability' explains much but misses cases (e.g. Travel Agents). The same 2019->2024 fit ACROSS CITIES has R2 of only 0.11 -- location, not just occupation mix, drove the spatial spread. (Figure 3)*

Large Dispersion ACROSS Firms in the Same Occupation-Industry

Figure 8: The Prevalence of Postings that Allow Hybrid or Fully-Remote Work Varies Greatly, even among Same-Industry Firms Recruiting in the Same Occupational Category



Note: Remote-work share of postings for firms in the SAME industry hiring in the SAME occupation. Aerospace management (A): Lockheed ~100% vs Boeing ~12% by 2025. Insurance math roles (B) and auto engineering (C) show similar splits (e.g. GM vs Honda swap places). Remote-work intensity is a firm CHOICE, not a fixed technological constraint. (Figure 8)

Takeaways & Contributions

- Methodological: fine-tuning a small, open-source LLM (DistilBERT) on domain text + human labels beats dictionaries, logistic regressions, and even frontier zero-shot AI models -- at tiny cost.
- Validation: the vacancy-based RWO measure correlates 0.70-0.93 with Census ACS/CPS remote-work shares across MSAs, states, and occupations.
- Empirical: advertised remote work rose 3x (US) to 5x+ (other countries) since 2019 and has NOT reverted -- a durable 'new normal'.
- Heterogeneity is deep and non-uniform: strongly persistent across occupations (R2=0.80) but weakly so across cities (R2=0.11), with wide gaps even between rival same-industry firms.
- Implication: remote-work suitability is an outcome of job design and firm choice, not a purely technological constraint -- data live and updated at WFHmap.com.